

Oluwatobi Melvyn Mayungbo

Data and Machine Learning Engineer

melvyn.t@gmail.com | [LinkedIn: linkedin.com/in/oluwatobi-mayungbo-3a567026b](https://www.linkedin.com/in/oluwatobi-mayungbo-3a567026b) | [GitHub: github.com/MelvTheGoat](https://github.com/MelvTheGoat)

SUMMARY

Data and machine learning engineer who builds automated data pipelines in Python. Built a forecasting platform with Prefect, dbt and DuckDB covering ingestion, validation, backtesting and drift monitoring, a reconciliation pipeline that matches multi-source feeds, and a crypto risk tool reading live on-chain market data.

SKILLS

Engineering and MLOps: Python, SQL, DuckDB, dbt-core, Prefect, Pytest, CI/CD (GitHub Actions), Docker, PostgreSQL, Redis, FastAPI, GCP Cloud Run, AWS ECS Fargate, Railway, MLflow, Evidently, Git, Streamlit, React

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

Payments and Risk: Payment Reconciliation, Web3 Fraud Risk Scoring, Transaction Fraud Detection, Credit Risk Decisioning, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

PROJECTS

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Engineered a multi-stage reconciliation pipeline using DuckDB and FastAPI, with temporal join logic and tolerance thresholds to handle asynchronous processor webhooks and bank settlement files.
- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Containerised and deployed the full service on Railway with structured log collection, automated integration testing and persistent transaction state tracking.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Built a resilient HTTP layer with caching, per-source rate limits and retries with exponential backoff and jitter; one failing source never breaks an investigation, and missing data lowers confidence rather than the score.
- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
 - Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER** May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

- B.Sc. Statistics, University of Ibadan** January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT** July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action